What Would Have Caught System A?

An incident review, and a competition to build something worse

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Duration: 45 to 60 minutes

Group size: 3

Output: Three sabotages, one Gate Sheet, and one entry in the Wall of Shame

Module context: Model Evaluation Metrics and Experiment Tracking. You have BLEU, chrF++, TER, ROUGE, and METEOR, and you have just been shown that all of them are the same thing wearing different hats. You do not yet have semantic metrics or judged evaluation. Everything here is answerable without them.

> **THE PREMISE** System A shipped. It ran for eleven days. A customer was told they were getting a refund and received a pressure washer instead. Every metric in your CI pipeline was green the entire time, and every one of them preferred System A to the correct output by roughly ten to one. You own the eval pipeline. Nobody is here to blame you. They are here to ask what you are going to do about it.

1. How this session runs

You will work in a group of three. Three is deliberate. Nobody gets to be a passenger, every position must be held by someone, and a group of three cannot hide a disagreement behind a majority. In your roles, you will walk through and respond to each round (as described below).

Attacking a system that you own is the fastest way to understand what it cannot see. It is also more fun than being told.

> **THERE ARE HINTS AND USING THEM IS NOT CHEATING** Section 12 has a stuck ladder for the five hardest questions, three levels each. Take one level at a time and stop as soon as you are moving again. Reading all three levels for a question you were about to solve is the only way to waste them.

2. Your role

Each person takes one role. Each one owns a different definition of the word caught, and the friction between those definitions is the discussion. Argue your role even when you privately agree with someone else.

Role	You own	The question you keep asking
Eval pipeline owner	The gate that was green for eleven days, and the CI budget	What check runs, on what, how often, and what does it cost per run?
Support operations lead	The agents reading these summaries, and every false alarm your check generates	How many of my people's hours does your check consume, and what do they do with a flag?
Risk and resourcing partner	The commitments made to customers in error, and what two engineers can finish before the freeze	Which of these failures do we have to answer for, and what are we giving up to prevent it?

3. The incident

Cordwell's agent-assist summarizer reads a support ticket thread and writes a one-sentence resolution summary into the agent queue. The agent reads the summary, then speaks to the customer. About 1,800 summaries are produced every working day.

System A was deployed on a Monday. It passed every gate. It ran for eleven days before a customer complaint escalated far enough that somebody read the outputs.

Quantity	Value
Summaries produced during the incident window	About 19,800 across 11 days
Summaries later found to state the wrong resolution type	About 1,228, or 6.2 percent
Cases where an agent repeated the wrong resolution to a customer	41
Formal complaints	3
Chargebacks	1
Cases referred to the product safety team	1

Quantity	Value
CI runs during the window that reported a regression	0

What the gate reported, the day System A shipped

```
Reference: "Customer received a cracked 2200 PSI pressure washer
            and wants a replacement unit shipped before Saturday."

System A:  "...and wants a refund issued before Saturday."      <- shipped
System B:  "The buyer's 2200 PSI power washer arrived damaged;
            they are asking for a replacement to arrive by Saturday."

  System              BLEU    chrF++     TER     R-1     R-2     R-L
  A  (wrong)         69.97     79.55   18.75   83.87   75.86   83.87
  B  (correct)        6.89     36.71   87.50   34.29   12.12   34.29
```

YOUR ASSIGNMENT What check, running before deploy, would have caught this, and what would it have cost? You will answer the second half with arithmetic, not adjectives.

4. The source ticket, CW-88214

This is the thread System A was summarizing. It is the ground truth. Read it carefully, because in Round 1 you are going to lie about it.

```
TICKET CW-88214           Opened Tue 09:14      Channel: web store
Customer: D. Whitfield     Product: CW-2200X pressure washer, 2200 PSI

[Tue 09:14  customer]
  Unit arrived this morning with a cracked pump housing and the
  GFCI plug is split open. I have a driveway sealing job booked
  Saturday and I need a working machine before then.

[Tue 09:41  agent M. Okafor]
  Do not plug the unit in. A split GFCI plug is a shock hazard.
  I have opened a replacement request for a CW-2200X.

[Tue 09:52  customer]
  Can I just get a refund instead and buy locally?

[Tue 10:06  agent M. Okafor]
  Refunds on damaged goods route through the returns team and take
  9 to 12 business days, which is after your Saturday job. I would
  not recommend it. Replacement ships faster.

[Tue 10:09  customer]
  Understood, go with the replacement then.

[Tue 10:15  agent M. Okafor]
  Replacement confirmed, arriving before Saturday. Return shipping
  is covered. The $34 expedite fee is not waived on this order.
```

NOTICE SOMETHING The word "refund" does appear in this ticket. So does the word "Saturday." So does a dollar figure. One naive check you are about to propose dies on this ticket, and it dies for an interesting reason.

5. Scoring by hand

You do not need a laptop for Round 1. ROUGE-1 F is computable on paper in under a minute, and the recipe below reproduces the library exactly. It was checked against the numbers on the previous page.

The recipe

- Lowercase both strings and split on anything that is not a letter or a digit. Punctuation disappears. Numbers count as tokens.
- Count matched tokens, with repeats capped. If "a" appears twice in the reference and three times in your candidate, that is two matches, not three.
- P equals matches divided by your candidate's token count. R equals matches divided by the reference's token count.
- F equals 2PR divided by (P plus R). Multiply by 100.

Worked, against the reference on the previous page

```
Reference (16 tokens):
  customer received a cracked 2200 psi pressure washer and wants
  a replacement unit shipped before saturday

System A (15 tokens):
  customer received a cracked 2200 psi pressure washer and wants
  a refund issued before saturday

  matched = 13     (everything except replacement, unit, shipped)
  P = 13/15 = 0.8667      R = 13/16 = 0.8125
  F = 2(0.8667)(0.8125) / (0.8667 + 0.8125) = 0.8387   ->   83.87

  The deck reports 83.87. So does the library. So does this recipe.
```

> **ONE CAVEAT, AND IT IS A CALLBACK** The library was run with use_stemmer=True. If your sabotage changes a word only by its ending, say washer to washers, the stemmer will match it and your hand count will not. Avoid ending-only changes, or note the discrepancy and claim the point anyway. This is Trap 1 from earlier today, arriving as a practical nuisance.

6. What things cost

Reference card for Round 2. All figures are for the Cordwell agent-assist pipeline at 1,800 summaries per working day. You will be asked to price a design, so treat these as the rate sheet.

Check	Build cost	Cost per run	Coverage it can afford
Surface metric in CI, as today	Already built	Milliseconds, free	100 percent
Deterministic extraction check: pull numbers, model codes, and dates from the summary and assert each appears in the source	About 1.5 engineer-days	Under a millisecond, free	100 percent

Check	Build cost	Cost per run	Coverage it can afford
Keyword presence check: assert the summary's resolution word appears in the source thread	About 0.5 engineer-days	Free	100 percent
A model reads the source and the summary and returns a verdict, running locally	About 3 engineer-days	About 2.4 seconds of one machine	100 percent costs 72 minutes of compute per day
The same, hosted	About 3 engineer-days	About $0.004 each	100 percent costs $7.20 per day, about $2,600 per year
Human reads the source and the summary	None	About 90 seconds each	100 percent is 45 hours per day. A 2 percent sample is 54 minutes per day.

What a flag costs downstream

If your check flags	That is per day	Which costs triage
3 percent of summaries	54 summaries	About 1.4 hours per day
8 percent of summaries	144 summaries	About 3.6 hours per day
15 percent of summaries	270 summaries	About 6.8 hours per day

You have a 200-ticket gold set with human-written summaries. You do not have a single labeled judgement of whether a summary is faithful to its source. Nobody has ever produced one.

7. Round 0: first instinct

Silent. Individual. Do not discuss.

Write three things, then stop.

- The check you would have added, in one sentence.
- What you think it costs, as a number, in whatever unit you like.
- One thing your check would still miss.

You will compare this against where you end up. Most people's Round 0 check turns out to be either much cheaper or much more expensive than they guessed.

8. Round 1: Break It

Group. This is a competition.

You are the adversary now. Your job is to write summaries of ticket CW-88214 that your own CI pipeline would wave through, and that would hurt somebody.

The rules

- Write three summaries. Each one must be a plausible thing a real model might actually emit. No gibberish, no keyword stuffing, nothing you would spot in a second.
- Each one must be factually wrong or unsupported with respect to the ticket on the previous page. Not merely awkward. Wrong.
- Each one must be a different kind of wrong. Two examples of the same trick count as one.
- Score each by hand using Section 5. The number to beat is System A at 83.87.
- For each one, write the harm in six words or fewer. Who gets hurt, and how.

Two kinds are already taken

You have seen both of these today, so neither counts as one of your three. They are here to show you what a kind is.

Kind	Already demonstrated by
Resolution swap	System A itself. The customer is promised a different outcome than the one agreed.
Polarity inversion	The drill bit minimal pair. Large becomes small, and BLEU and ROUGE-1 score both identically.

1.1 Build your three. Fill in the table in Section 10, Part 1, as you go.

1.2 For each sabotage, say in one line why the metric did not care. Point at the mechanism: clipping, the brevity penalty, unigram overlap, the stemmer, the fact that nothing read the ticket. "Because it is dumb" is not a mechanism.

1.3 Which of your three was hardest to make score highly, and what does that tell you? There is at least one kind of dangerous wrongness that surface metrics are accidentally decent at catching, and finding it is worth more than another 90-scorer.

1.4 Name your three kinds. Give each one a short label you could put in a bug tracker. You are building a taxonomy, and you will need it in Round 2, because a check that catches one kind may be useless against another.

9. Round 2: the review

Group. Fill in Section 10, Part 2.

You have just built multiple kinds of dangerous wrongness between you, all of which your pipeline endorses. Now you are the eval owner again.

2.1 Go through your three sabotages plus the two that were already taken. For each one, name a check that would catch it. Then sort your checks into two piles: the ones that need to read the source ticket, and the ones that do not. Which pile is bigger, and what does that tell you about where your pipeline's blind spot actually lives?

2.2 Not every check that reads the source is expensive. Using the cost card in Section 6, identify the cheapest check on your list that would have caught the real System A, and price it: build cost, run cost, and the triage cost of its false alarms. Then say what fraction of your five kinds it catches.

2.3 Somebody proposes the obvious fix: collect more reference summaries, three or four per ticket instead of one, so paraphrases like System B stop being punished. Price it, using the 200-ticket gold set, then answer the harder question. When BLEU has several references, an n-gram in your candidate counts as matched if it appears in any one of them. Given that rule, what happens to System A's score as you add references? Work out the direction before you argue about the cost.

Do not skip the mechanism. The direction of the effect surprises most people, and it settles the question far more decisively than the budget does.

2.4 Take the check you priced in 2.2 and try to break it, using ticket CW-88214 specifically. There is at least one sabotage on this ticket that passes it cleanly, and the ticket was written to make that possible. Find it, and say what property of the ticket lets it through.

2.5 Design the gate. Not one check, a sequence. For each layer say what it runs on, how often, what it costs per day using Section 6, and what it hands to the next layer. Your total human cost must fit inside four hours per day, and you must be able to state the flag rate you are assuming.

2.6 Name what your gate still misses. Be specific enough that somebody could build the sabotage that gets through it. If your group cannot name one, you have not finished designing, you have finished hoping.

2.7 The counterfactual question, and answer it with a number. Had your gate been running the day System A shipped, how many of the 19,800 summaries would have been flagged, how many of the 41 wrong customer commitments would have been prevented, and how long would it have taken to notice? An honest "we would have caught it on day one, and here is the flag rate we would have paid for that" is the target.

10. Your deliverable: the Gate Sheet

Part 1. Your sabotages

#	The summary you wrote	ROUGE-1 F, by hand	The kind	The harm, six words
1				
2				
3				

Part 2. The gate

Layer	What it runs on, and how often	Cost per day	Which kinds it catches, and which it does not
1			
2			
3			

Total human cost per day: ________ **Assumed flag rate:** ________ **What it still misses:** ________________________

11. Round 4: the Wall of Shame

Read your best sabotage out loud, with its score. Then, in one breath each:

#	State this	What makes it count
1	The sabotage and its ROUGE-1	Read the summary verbatim. Say the number. Let the room react.
2	The kind of wrongness it is	The bug-tracker label from question 1.4.
3	The cheapest thing that catches it, with its cost per day	A number from Section 6. "A judge" without a cost is not an answer.
4	What your gate still misses	Specific enough that somebody could build it.

The two awards

Award	Vote for the sabotage that...
Most Dangerous	would cause the most harm if it shipped
Most Likely to Actually Ship	a real model would plausibly emit, that a hurried reviewer would wave through, and that nobody would notice for a week

The second award is the one that matters. Spectacular failures get caught. The dangerous ones are boring, plausible, and one word different from correct.

12. If your group is stuck

Three levels per question. Take one at a time and stop as soon as you are moving again. None of these contains an answer and reading all three when level one would have done is the only way to waste them.

Question 1.1 and 1.2 | Getting a sabotage to score high

Level 1. Look at the arithmetic in Section 5. Your score is driven by how many tokens you leave alone. So the question is not what to write, it is how little you can change.

Level 2. System A changed three tokens out of sixteen and still scored 83.87. What could you change by exactly one token that would matter more than System A's change did? Scan ticket CW-88214 for tokens where a single substitution changes an obligation, a quantity, or a person.

Level 3. There are two shapes worth trying. Substitute one token so the sentence stays the same length: the score barely moves and the meaning can invert completely. Or add one or two tokens rather than replacing any: nothing that matched stops matching, your recall is untouched, and precision falls only slightly. That second shape is how you promise something the ticket never granted.

Question 1.4 | Naming the kinds

Level 1. A kind is defined by what a check would have to do to catch it, not by what the sentence says. Two sabotages are the same kind if the same check catches both.

Level 2. Sort your sabotages by asking: could a program with no understanding of English catch this, if it were allowed to read the ticket? Some of yours are a string comparison away from being caught. Others are not.

Level 3. One useful split is between claims that can be checked against a closed set of things present in the source, such as numbers, dates, model codes, and named parties, and claims about relationships, such as who owes what, whether something was agreed or declined, and what was promised. The first group is mechanically checkable. The second needs something that understands the thread. Your taxonomy should make that boundary visible.

Question 2.2 | The cheapest check that catches System A

Level 1. System A's error was the word refund. Ask the simplest possible question about that word and the source ticket.

Level 2. Before you commit, go and look at what ticket CW-88214 actually contains. Read the timestamps at 09:52 and 10:06. Then ask whether your simple question returns the answer you were hoping for.

Level 3. Presence is not support. A word can appear in a source and still make the claim false, because the source may be declining it, quoting it, or discussing it hypothetically. So a presence check is nearly free, catches a real class of errors, and is defeated by this specific ticket. Both halves of that sentence are worth saying out loud, and question 2.4 is where you say the second half.

Question 2.3 | What more references actually do

Level 1. Write down the multi-reference rule first: an n-gram counts as matched if it appears in any one of the references. Then ask what adding a reference can do to a count computed that way.

Level 2. A count that takes the best available match can only stay the same or go up when you add another option. Apply that to System A specifically, and then to System B, and compare what happens to the gap between them.

Level 3. More references loosens the match rather than tightening it, so the factually wrong system's score does not fall and can rise. The gap between wrong and correct narrows by a little, not by enough. The finding is not that references are worthless; it is that they are the wrong axis, because no number of reference summaries introduces anything that reads the ticket. That is why this cannot be the answer to the incident.

Question 2.5 | Designing the gate

Level 1. You cannot afford one check that does everything. So stop looking for one and start ordering the ones you have by cost.

Level 2. Section 6 gives you three price points that differ by orders of magnitude. Anything free can run on everything. Anything costing seconds can run on everything if you are patient. Anything costing 90 seconds of a person cannot run on more than a sliver.

Level 3. Put the free deterministic checks first and let them run on 100 percent, because they have high precision on the kinds they cover and they cost nothing. Let the expensive check run on what survives, or on a sample. Reserve human time for keeping the layer above it honest rather than for reviewing output directly. Then state the flag rate you are assuming, because that assumption is what your whole budget rests on.

13. Vocabulary for this discussion

Term	Working definition in this context
Reference-based	The metric compares your output to a human-written answer. Every metric in your CI pipeline today is reference-based, which means none of them has ever read a ticket.
Grounded in the source	Every claim in the output is supported by the input document. Distinct from matching a reference, and distinct from being fluent.
Presence versus support	A word appearing in the source does not mean the source supports a claim about it. Sources decline things, quote things, and rule things out.
Closed-world claim	A claim checkable against a finite set of things present in the source: numbers, dates, model codes, named parties. Mechanically verifiable, cheaply.
Relational claim	A claim about who owes what, what was agreed, what was declined, what was promised. Not mechanically verifiable from token presence.
Flag rate	The fraction of outputs your check sends to a human. It is the number that decides whether your design is affordable, and it is usually the number nobody estimates.
Precision of a check	Of the things your check flags, how many were really wrong. A check with poor precision does not fail loudly, it fails by exhausting the people who answer it.
Tripwire	A cheap check that detects that something changed, without telling you whether the change was good. Surface metrics are tripwires. They were doing their job during the incident; nothing changed in the surface statistics, because nothing did.